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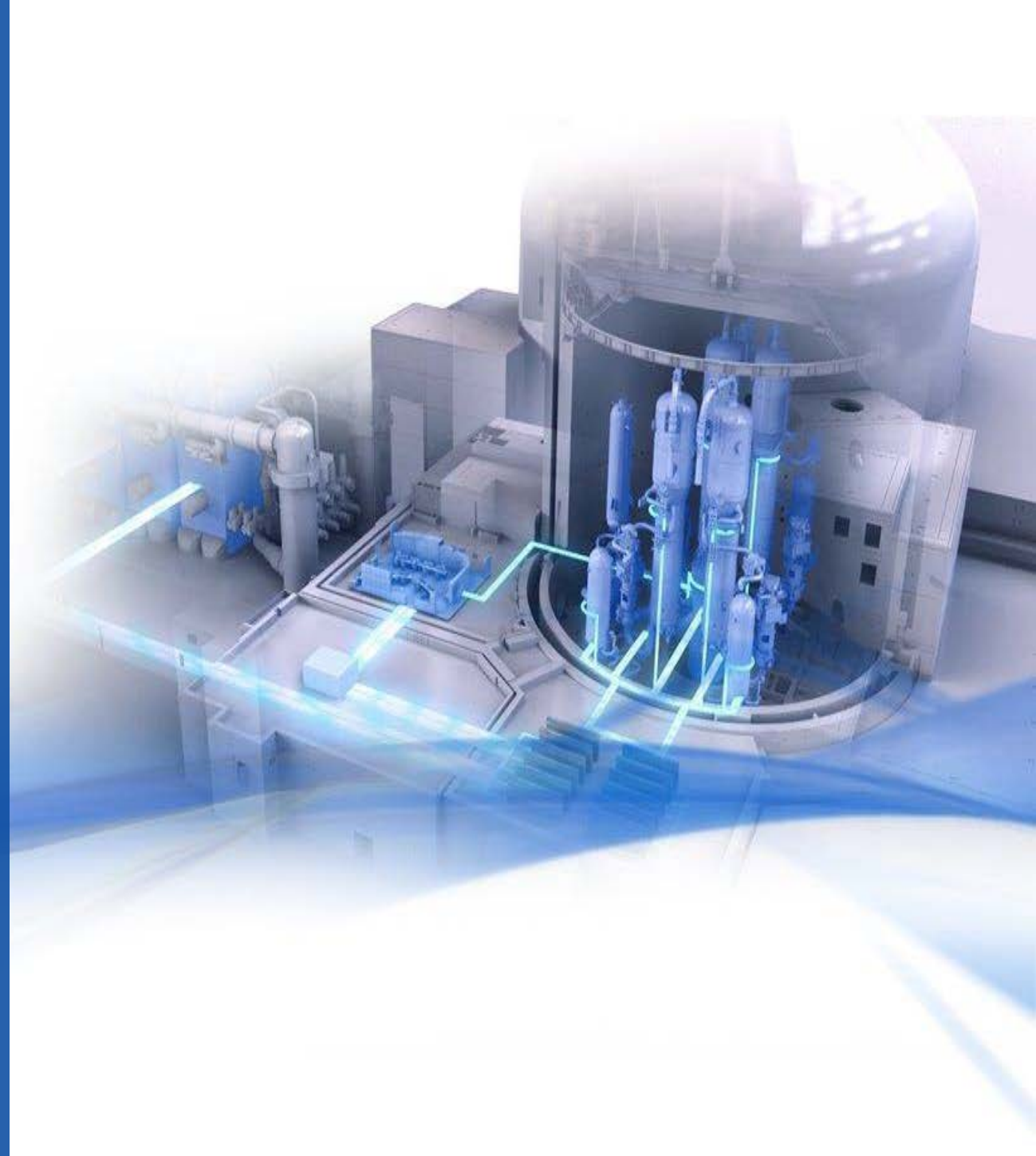
INSTRUMENTATION AND CONTROL

Safety I&C Modernizations

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OUTLINE

1. Introduction
2. References
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INTRODUCTION: Safety I&C Modernization

Challenge

Equip nuclear plants with updated technologies and improved processes to deliver greater economic efficiency and reliability

Objective

- High reliability & availability
- Reducing operating and maintenance costs
- Long-term support
- State-of-the-art technology

Key to success

- Outstanding references
- Comprehensive expertise: Engineering, integration and project management
- Proven platforms

MODERNIZATION REFERENCES

USA



- Browns Ferry 1-3
- Oconee 1-3
- Sequoyah 1&2
- St. Lucie
- Watts Bar 1&2

CHINA



- Qinshan 1
- Ling Ao
- Daya Bay

CANADA



- Bruce A 1&2
- Darlington 1-4
- Point Lepreau

BRASIL



ANGRA 2

ARGENTINA



Atucha 1, 2

SOUTH AFRICA



Koeberg 1&2

MODERNIZATION REFERENCES



COMPREHENSIVE EXPERTISE

PLATFORM

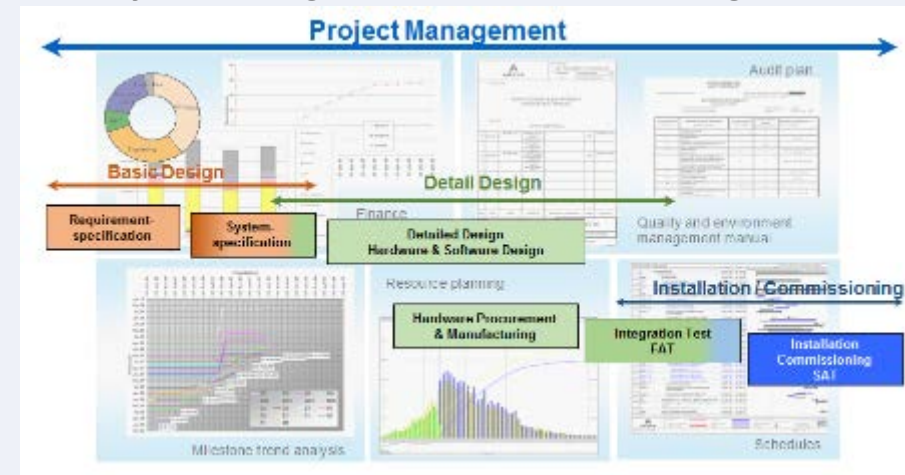
Continuous Management of Product Lifecycle and Obsolescence Mitigation

- In order to support the TXS installed fleet in the long term, Framatome has developed upgrade solutions allowing the use of TXS GEN2 components in TXS GEN1 applications.
 - Processing Unit
 - I/Os
 - Communication
 - Power Supply and Monitoring
- The Upgrade of the TXS Core Software doesn't require the exchange of the TXS Hardware. The upgrade requires the change of
 - Service Unit and Gateway
 - Engineering and Design Unit

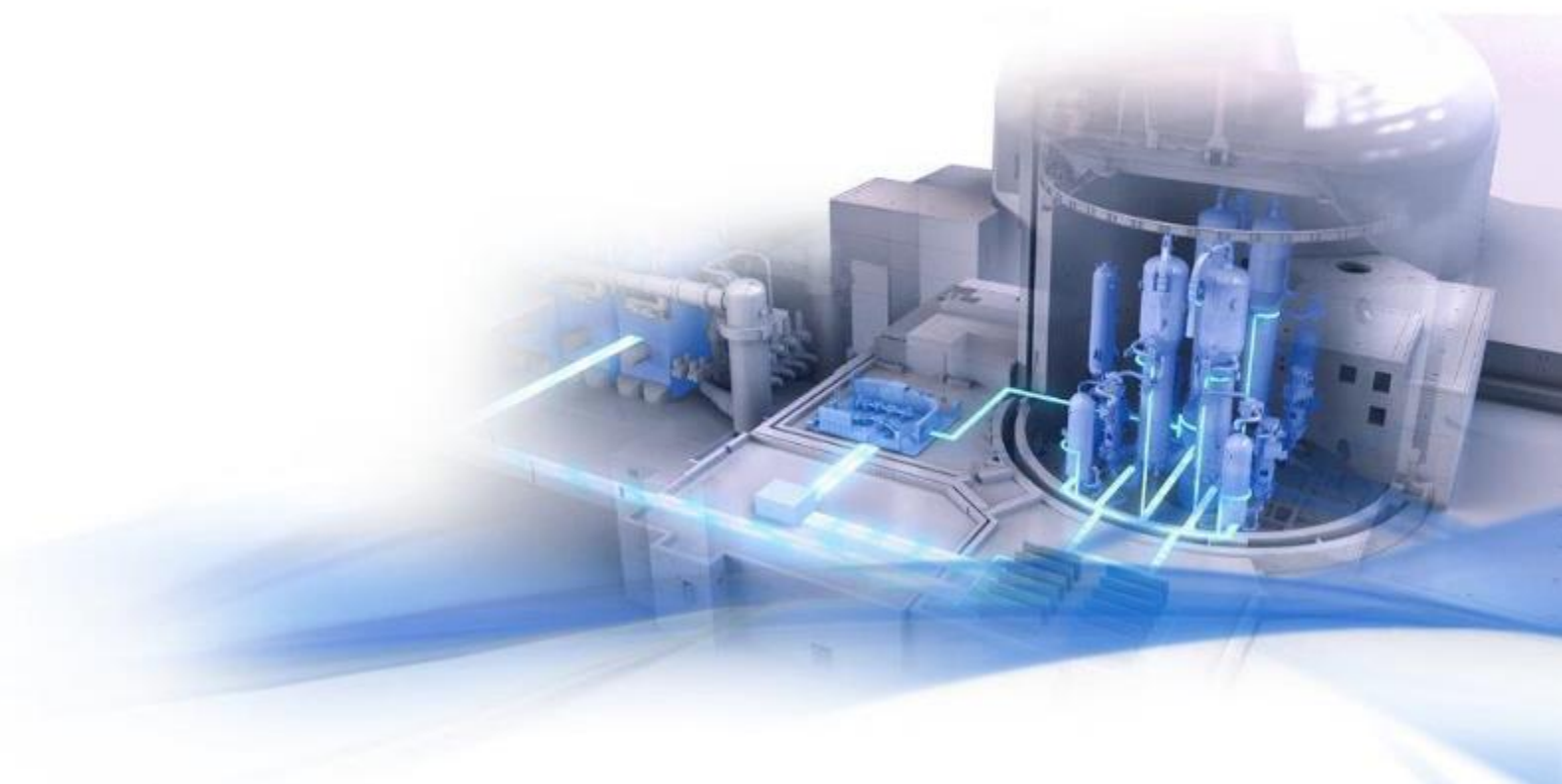
PROJECT

Project Management, Design & Integration

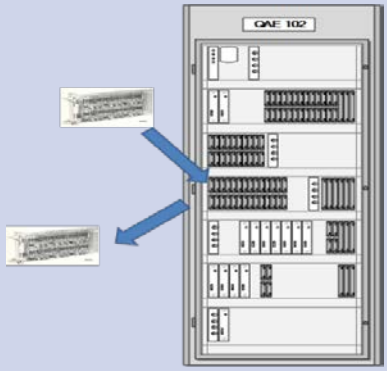
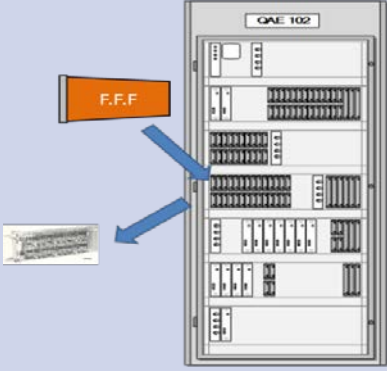
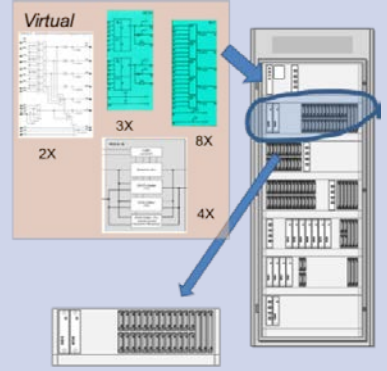
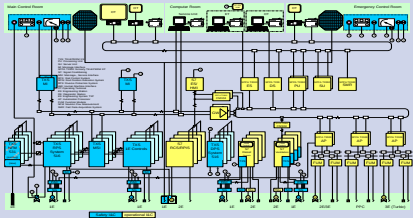
- Well defined and proven processes
- Integration of NON-FRA Products
- Experienced specialist in all competence areas
 - Requirement & project management
 - ...
 - System integration and commissioning



MODERNIZATION PROCESS



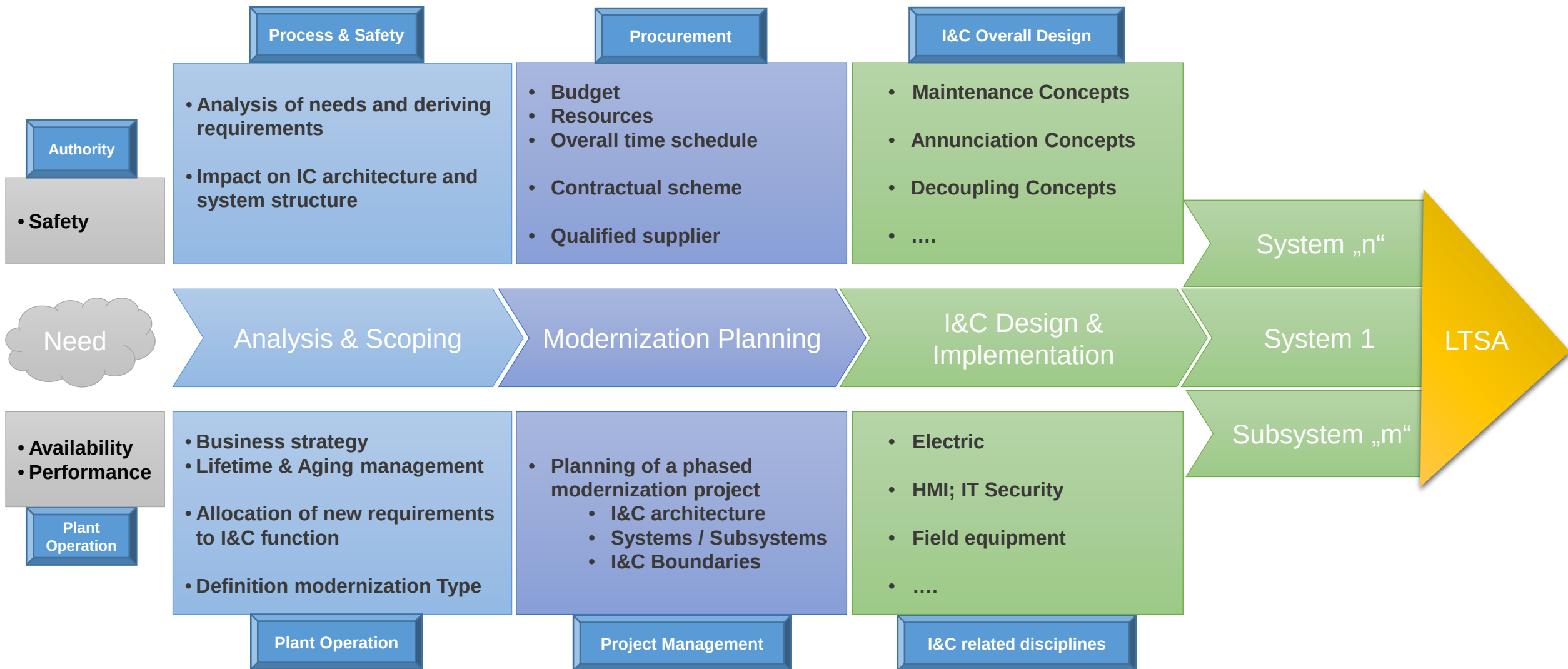
TYPES OF I&C MODERNIZATION

(1) Spare part	(2) Form, fit and function (FFF)	(3) Rack replacement	(4) I&C System modernization	(5) Comprehensive modernization
				
• Cost effective in short time • Fast replacement during outage • No effect on documentation	• Reverse engineering limited on module level • Fast replacement • Small effect on documentation	• Equipment frame treated as black-box • Similar interface within cabinet • Use of existing I&C products / platform	• Same footprint • Replacement during a single outage • Revolve safety deficiencies • Improve operation	• Cost effective in the long term (lifetime extension program) • Implementation of complex safety deficiencies
• Failure rate & spare part costs growing • Dependability of few skilled people • Limit availability of components	• New product development + Qualification • Lack of knowledge on installed products • High investment on clone product • OEM intellectual properties	• Reverse engineering full equipment frame • Medium affect on documentation • Management of interfaces	• More complex redesign • Reverse engineering for new I&C system • New documentation • New licensing • Cybersecurity	• Complex replacement + reverse engineering for whole plant • Challenge on regulatory approval • Extensive quantity of stakeholders

PRO

CON

I&C MODERNIZATION PROCESS



MODERNIZATION EXAMPLES AND PLATFORM STRATEGY: TELEPERM XS

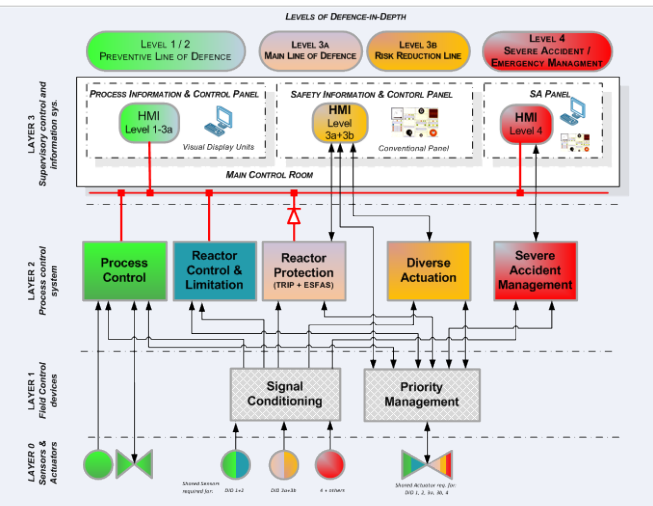


TELEPERM XS

Comprehensive platform for safety solutions

TELEPERM XS for meeting highest safety requirements

- Reactor Protection System
- Reactor Control and Limitation
- Core Monitoring and Control
- Neutron Flux Measurement
- Control Rod Controls
- ...

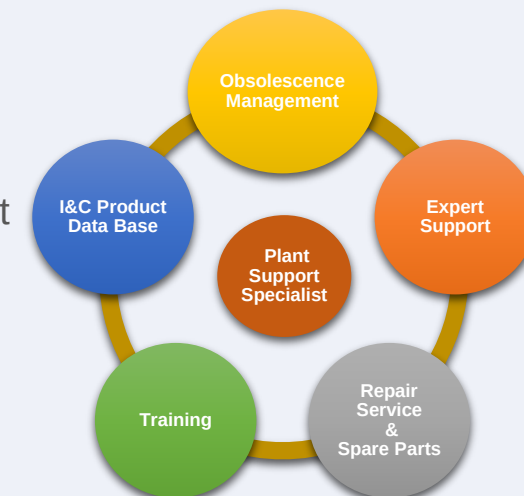


TELEPERM XS generic evidences are updated continuously

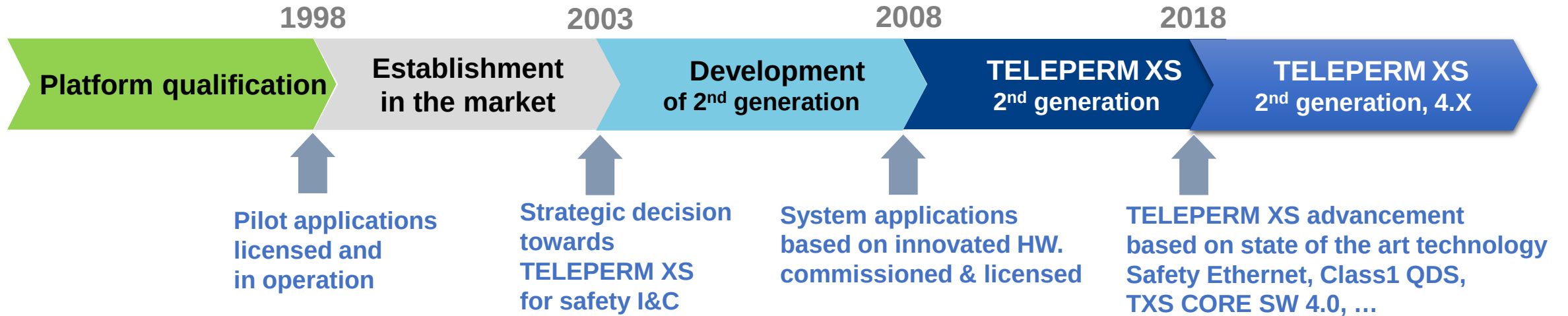
- Requirements on application-oriented languages & code generation
- Requirements on firmware and complex electronic components
- Climatic testing, mechanical load, EMC compatibility
- Cyber Security

Lifecycle Solutions

- Outage support
- Maintenance
- Obsolescence management
- Spare parts
- Repair & refurbishment
- Reverse engineering
- Training



TELEPERM XS: PROVEN PLATFORM



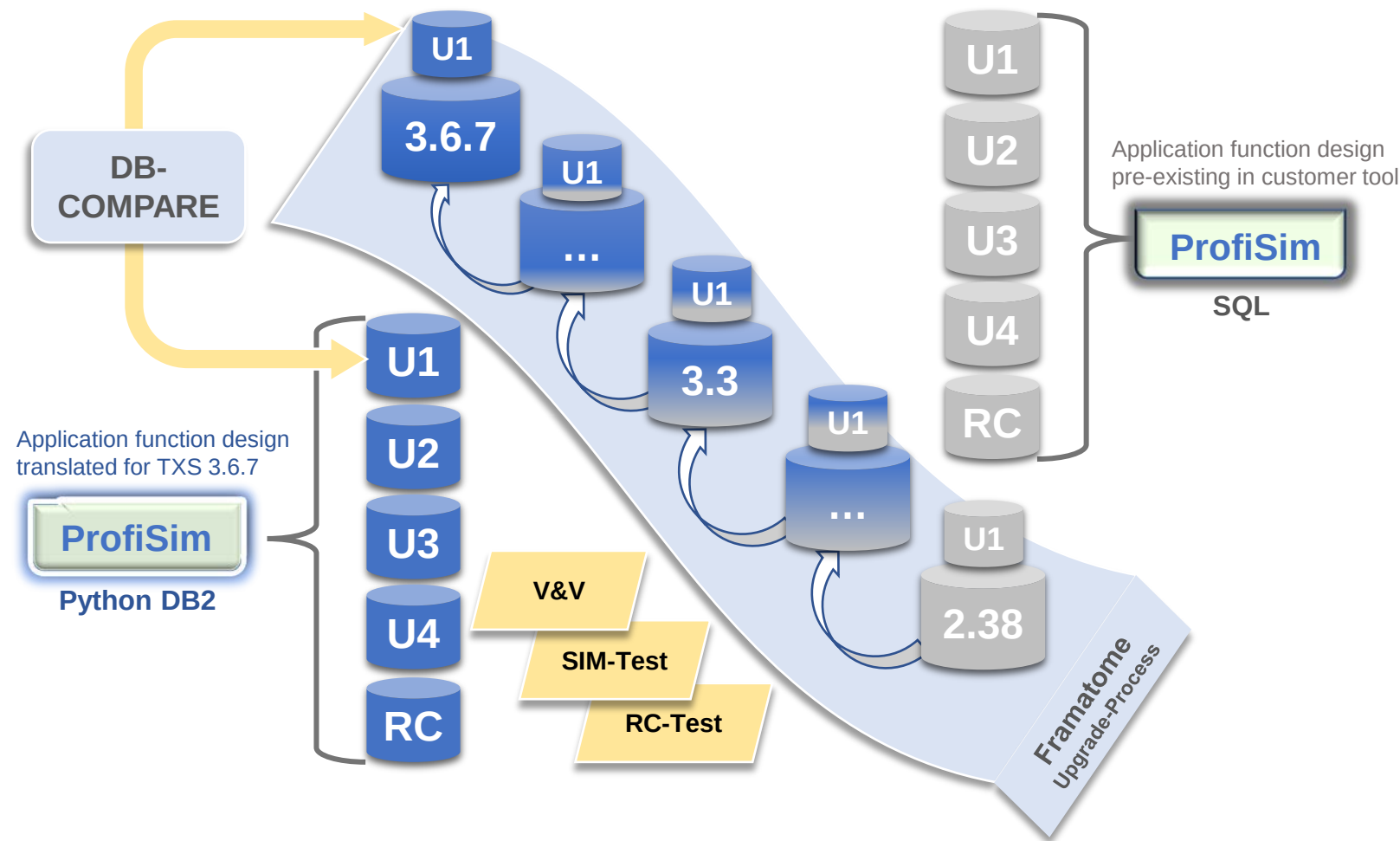
- 2 decades of experience gained in digital safety I&C life cycle management
- Evolutionary development approach continued taking into account next projects requirements and long-term commitments
- TELEPERM XS platform and qualification is continuously maintained, and proven by successfully licensed applications in 17 countries

MODERNIZATION: PLATFORM OVERVIEW

	TXS 1G	TXS 2G 3.x	TXS 2G 4.x
Processing	SVE1	SVE2 SVE2+	SVE3
Input/Output	S4x S7x	SAx SDx SGPIO	SAx+ SDx+ SGPIO+
Communication	SL21 SLLM SCP1 SHO1	SL22 SLM2 SCP3 SES1	SEI1 SES2 SSW1 SMC1
Power Supply	SES72	SCxU	PSU1
Conditioning	SSI1;...	SAA1; SNV1; STT1; SBC1...	SCxI SCU SCCI
Priority Control	AV42x		
Core -SW	TXS 2.38	TXS 3.0 TXS 3.3 - TXS 3.6	TXS4.0 - TXS 4.x

EXAMPLE: TXS CORE SOFTWARE UPGRADE

Integrated Project Team: MVM Paks NPP (Customer) + Framatome / Evopro



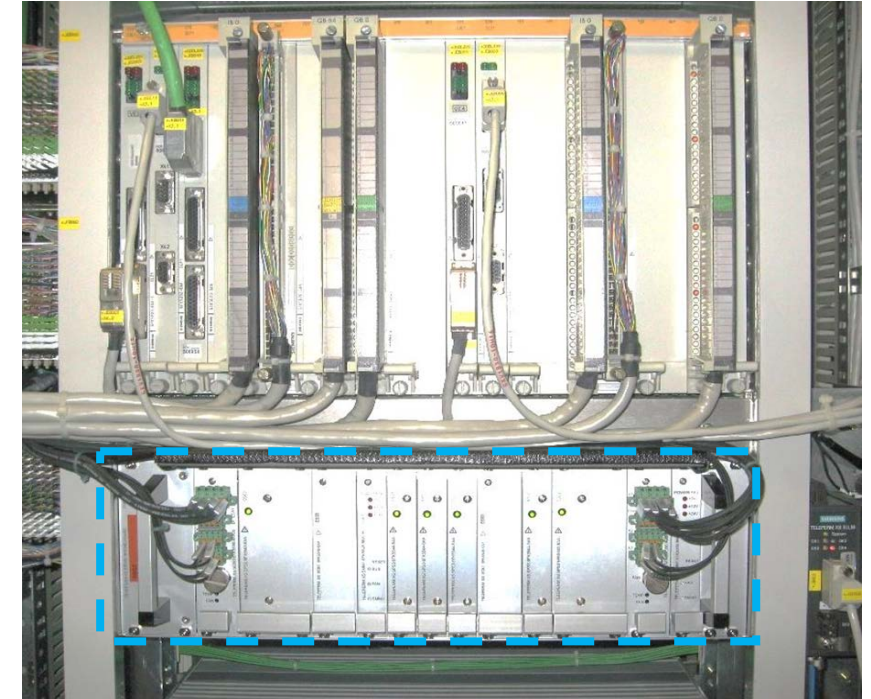
Engineering, V&V	
F:	TXS-DB upgrade from 2.38A to 3.6.7
F:	PyDBv2 interface (TXS 3.6.7)
F:	Hardening, time sync of SU/GW
F:	Hardware design
C:	Automated DB-Gen using PyDBv2
C:	Simulator upgrade
C:	Test machine upgrade
Services on Site	
F:	Commissioning support
F:	Installation of SU/GW
C:	Commissioning

Securing Know-how

EXAMPLE: HARDWARE UPGRADE

Upgrade of Power Supplies TXS Gen1 → Gen2

- Latest version of TELEPERM XS power supplies installed
- No removal of subrack needed
- Plug-in solution (SBGA1/2 adapter)
- No reverse engineering required
- No change of software required



References:

- The upgrade of the SBG1/2 sub rack power supplies was successfully performed in RPS system in Beznau (Switzerland), and applications in Grohnde (Germany), in Kozloduy (Bulgaria) and Parks (Ungarn).

CABINET POWER SUPPLY & MONITORING

TXS PSU 1 power supply & monitoring unit

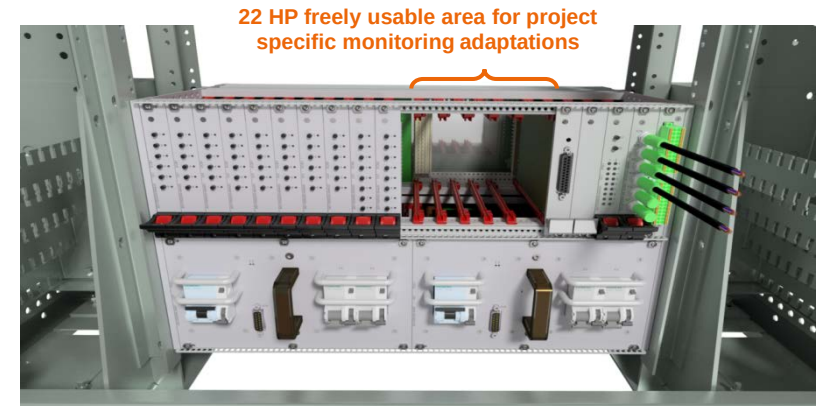
- PSD1 Power Supply Diode
 - replaces SED31-1 and Cabinet EMC Filter Capacitors
- Up to 8 ECB1 Electronic Circuit Breaker
 - replaces VA63 (6FP1707-0B)
- Up to 2 ECB2 Electronic Circuit Breaker
 - replaces SS11
- CMU1 Cabinet Monitoring Unit
- BSU1 Backplane Supply Unit
 - replaces wiring

Main benefits

- Saves 2 height units cabinet space
 - space consumption supply and monitoring: 6U 84HP
- Much better airflow compared to previous PS variants
- Operational temperature range up to 70 °C without derating
- Possible use in new cabinet TXS 4.x
 - and type SC9422 TXS 2G
 - and type STS24-1 TXS 1G



SES72-24 mounted in STS24-1



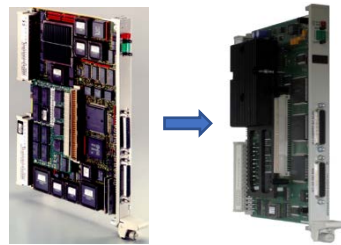
PSU1 Variant „UPG-1G“ mounted in STS24-1

TELEPERM XS STRATEGY: 1GEN → 2GEN

Examples

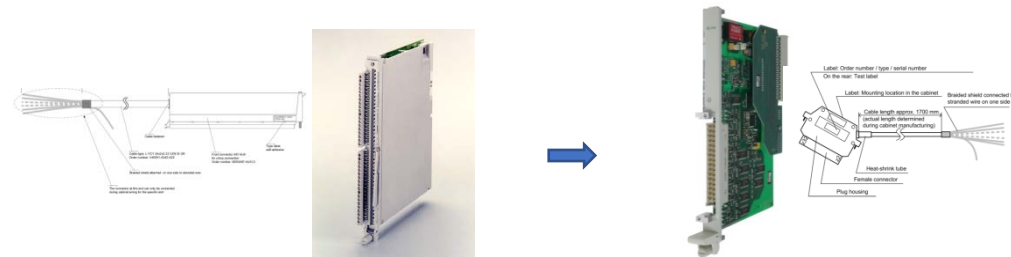
Processing

- SVE2 module can replace the SVE1 in-situ of the SBG1/2 sub rack
- SVE2 requires two slots cause of larger heat sink
- SVE1 and SVE2 shall not be mixed within one SBG1/2
- The SK-SVE front cable can be re-used.
- At least TXS Core SW 3.3.x



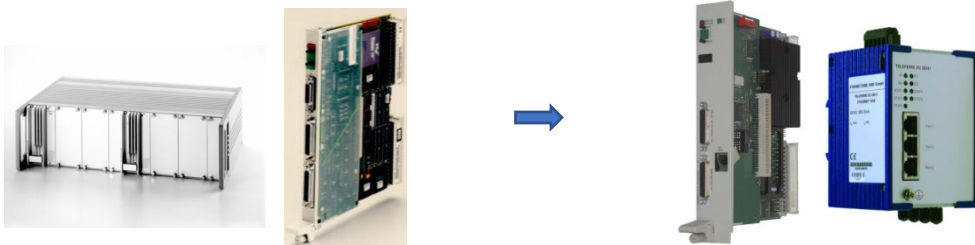
Input / Output

- When replacing the I/O boards, the respective front connector cables have to be replaced as well.
- I/O modules can operate in mixed configuration
- No changes are needed for the current application SW



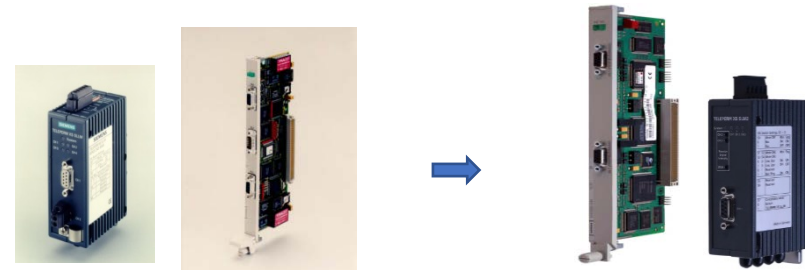
Communication, Ethernet H1

- The upgrade of the Ethernet network requires certain mechanical and electrical modification
- The SHS1+SHO2+SHT2 will be replaced by the SES2
- The SCP3 requires two slots cause of larger heat sink
- At least TXS Core SW 3.3.x



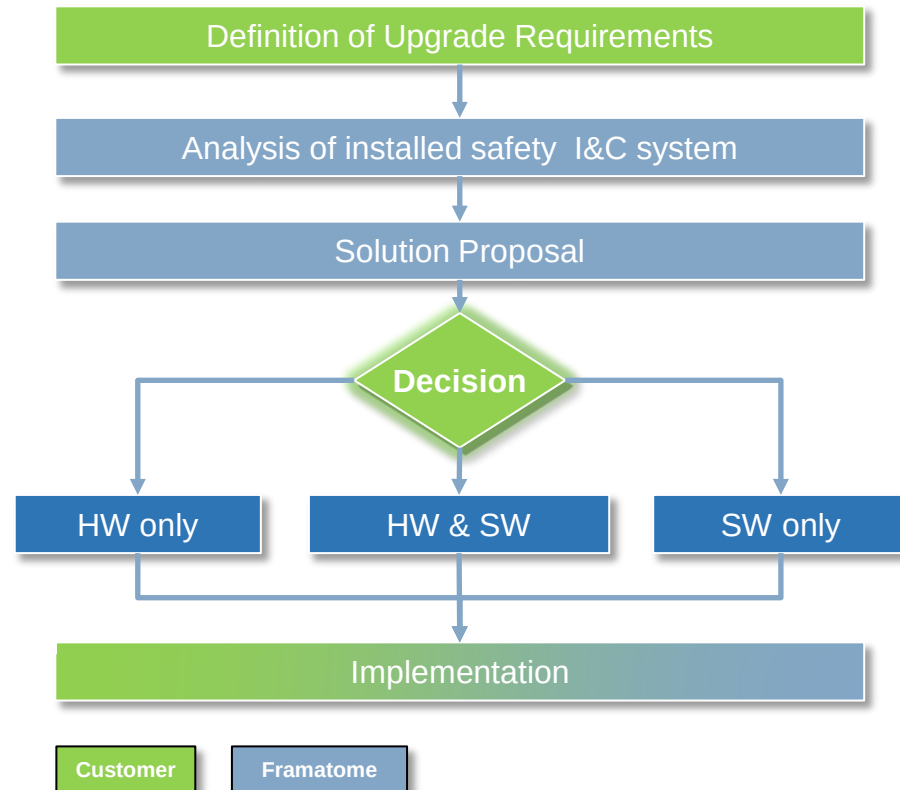
Communication, Profibus L2

- Within one L2 network, mixed configurations are not possible
- the new communication modules have to be engineered and new application code has to be generated and uploaded
- At least TXS Core SW 3.3.x



UPGRADE APPROACHES

Driven by need, in close cooperation with customers

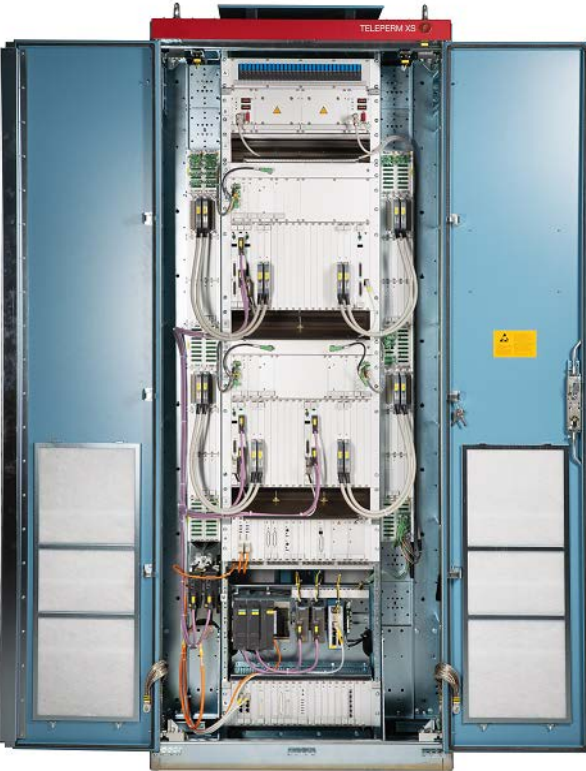


- Extension of features
- Need for a standardized interface to other plant systems
- Preparation for usage of latest TELEPERM XS hardware
- Benefit of enhanced functions provided by latest SW release
- Replacing of GW and SU Hardware
- Securing Know-How for the next generation of engineers

**Planning, design and implementation:
Ensuring effective integration and licensing**

OUR COMMITMENT

- Delivering integrated and innovative solutions for new builds as well as lifetime extension and modernization projects for nuclear power plants



Full range of packaging options

- Standard cabinets
- Wall-mounted enclosure
- Mixes configuration 3U/6U in sub racks

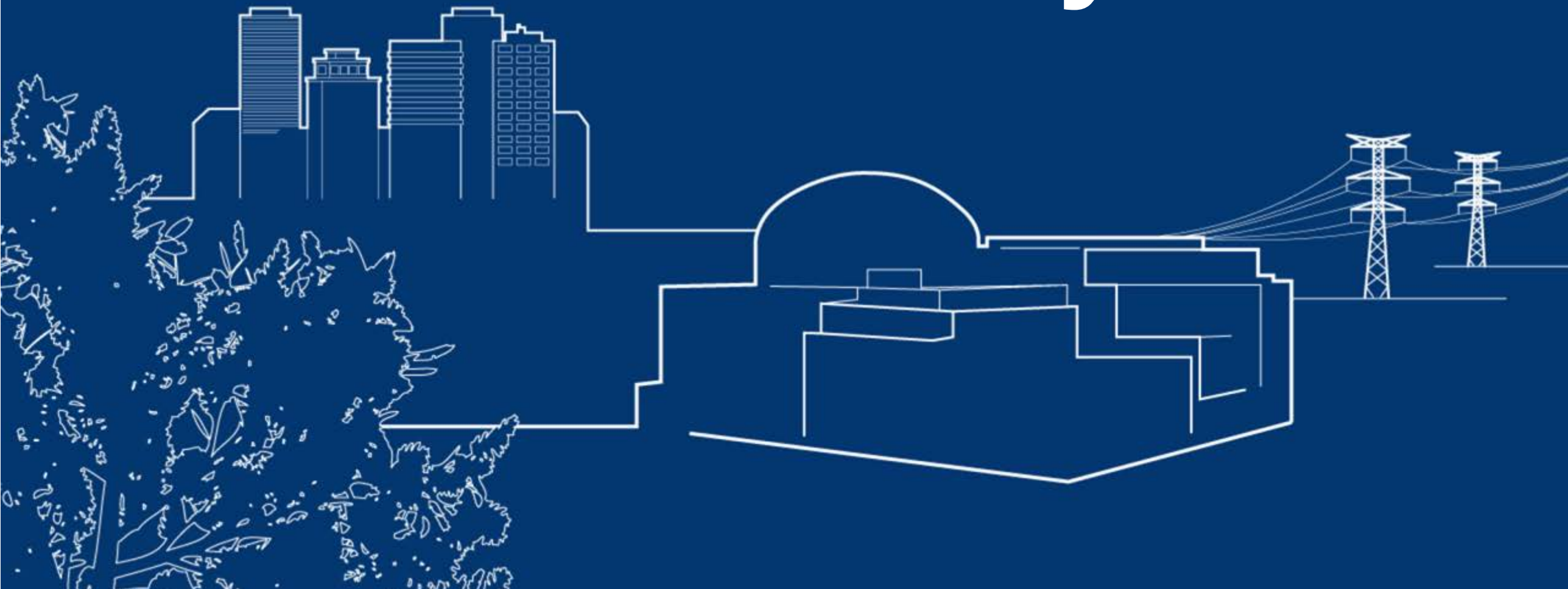
Modules to fit all needs

- Analogue & digital I/O modules
- Signal conditioning
- Processing modules
- Neutron Monitoring modules
- Amplifiers, HV supply, test signal generation ,...
- Auxiliary modules (fusing, isolating, ...)
- Power Supply

**Your performance
is our everyday commitment**

framato**me**

Thank you



CUSTOMER BENEFITS AT A GLANCE



Requirements, Specifications, Architecture & Systems Design

Requirements engineering, tool based documentation



State-of-the art technology

Digital and non-digital solutions



Low maintenance cost

High reliability & availability
Self-monitoring, Sustainability

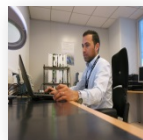


Improved Safety

Qualified for all safety & process purposes



High performance and advanced controls



Project Management, HMI+HFE, Cybersecurity, V&V Services, Installation & Commissioning



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